

i.e., $\varphi(s^*) \leq \varphi(s)$ for all $s \in \mathcal{S}$, and so φ has property (c).

To show properties (a) and (b), let $v \in \mathcal{F}$ with $\|v\| < 1$. Then, $u = \eta v + f$ lies in $\mathcal{B}_\eta(f)$. With (3.30) and by the linearity of φ , we have

$$\varphi(v) = \varphi\left(\frac{u - f}{\|s^* - f\|}\right) < \varphi\left(\frac{s^* - f}{\|s^* - f\|}\right).$$

This implies

$$\|\varphi\| = \sup_{\|v\| \leq 1} |\varphi(v)| \leq \varphi\left(\frac{s^* - f}{\|s^* - f\|}\right)$$

and, moreover, by using the continuity of φ once more, we have

$$\|\varphi\| = \frac{\varphi(s^* - f)}{\|s^* - f\|} \iff \varphi(s^* - f) = \|\varphi\| \cdot \|s^* - f\|.$$

If we finally normalize the length of $\varphi \neq 0$, by scaling φ to unit norm, i.e., $\|\varphi\| = 1$, then φ satisfies properties (a) and (b). $\blacksquare$

3.4 Direct Characterization

In this section, we develop necessary and sufficient conditions for the minimization of convex functionals. We then apply these conditions to norms to obtain useful characterizations for best approximations. To this end, we work with *Gâteaux¹³ derivatives* to compute directional derivatives for relevant norms. In the following discussion, $\mathcal{F}$ denotes a linear space.

Definition 3.49. For a functional $\varphi : \mathcal{F} \rightarrow \mathbb{R}$,

$$\varphi'_+(u, v) := \lim_{h \searrow 0} \frac{1}{h} (\varphi(u + hv) - \varphi(u)) \quad \text{for } u, v \in \mathcal{F} \quad (3.31)$$

is said to be the **Gâteaux derivative** of φ at u in direction v , provided that the limit on the right hand side in (3.31) exists. $\circ$

For convex functionals $\varphi : \mathcal{F} \rightarrow \mathbb{R}$ we can show that the limit on the right hand side in (3.31) exists.

Theorem 3.50. Let $\varphi : \mathcal{F} \rightarrow \mathbb{R}$ be a convex functional. Then, the Gâteaux derivative $\varphi'_+(u, v)$ exists for all $u, v \in \mathcal{F}$. Moreover, the inequality

$$-\varphi'_+(u, -v) \leq \varphi'_+(u, v) \quad \text{for all } u, v \in \mathcal{F}$$

holds.

¹³ RENÉ GÂTEAUX (1889-1914), French mathematician

Proof. Let $\varphi : \mathcal{F} \rightarrow \mathbb{R}$ be a convex functional. We show that for any $u, v \in \mathcal{F}$ the difference quotient $D_{u,v} : (0, \infty) \rightarrow \mathbb{R}$, defined as

$$D_{u,v}(h) = \frac{1}{h} (\varphi(u + hv) - \varphi(u)) \quad \text{for } h > 0, \quad (3.32)$$

is a monotonically increasing function in $h > 0$, which, moreover, is bounded below. To verify the monotonicity, we regard the convex combination

$$u + h_1 v = \frac{h_2 - h_1}{h_2} u + \frac{h_1}{h_2} (u + h_2 v) \quad \text{for } h_2 > h_1 > 0.$$

The convexity of φ then implies the inequality

$$\varphi(u + h_1 v) \leq \frac{h_2 - h_1}{h_2} \varphi(u) + \frac{h_1}{h_2} \varphi(u + h_2 v)$$

and, after elementary calculations, the monotonicity

$$D_{u,v}(h_1) = \frac{1}{h_1} (\varphi(u + h_1 v) - \varphi(u)) \leq \frac{1}{h_2} (\varphi(u + h_2 v) - \varphi(u)) = D_{u,v}(h_2).$$

If we now form the convex combination

$$u = \frac{h_2}{h_1 + h_2} (u - h_1 v) + \frac{h_1}{h_1 + h_2} (u + h_2 v) \quad \text{for } h_1, h_2 > 0,$$

we obtain, by using the convexity of φ , the inequality

$$\varphi(u) \leq \frac{h_2}{h_1 + h_2} \varphi(u - h_1 v) + \frac{h_1}{h_1 + h_2} \varphi(u + h_2 v)$$

and, after elementary calculations, we obtain the estimate

$$\begin{aligned} -D_{u,-v}(h_1) &= -\frac{1}{h_1} (\varphi(u - h_1 v) - \varphi(u)) \\ &\leq \frac{1}{h_2} (\varphi(u + h_2 v) - \varphi(u)) = D_{u,v}(h_2). \end{aligned} \quad (3.33)$$

This implies that the monotonically increasing difference quotient $D_{u,v}$ is bounded from below for all $u, v \in \mathcal{F}$. In particular, $D_{u,-v}$ is a monotonically increasing function that is bounded from below. Therefore, the Gâteaux derivatives $\varphi'_+(u, v)$ and $\varphi'_+(u, -v)$ exist. By (3.33), we finally have

$$-\frac{1}{h} (\varphi(u - hv) - \varphi(u)) \leq -\varphi'_+(u, -v) \leq \varphi'_+(u, v) \leq \frac{1}{h} (\varphi(u + hv) - \varphi(u))$$

for all $h > 0$, as stated. ■

Now we note a few elementary properties of the Gâteaux derivative.

Theorem 3.51. *Let $\varphi : \mathcal{F} \rightarrow \mathbb{R}$ be a convex functional. Then the Gâteaux derivative φ'_+ of φ has for all $u, v, w \in \mathcal{F}$ the following properties.*

- (a) $\varphi'_+(u, \alpha v) = \alpha \varphi'_+(u, v)$ for all $\alpha \geq 0$.
- (b) $\varphi'_+(u, v + w) \leq \varphi'_+(u, v) + \varphi'_+(u, w)$.
- (c) $\varphi'_+(u, \cdot) : \mathcal{F} \rightarrow \mathbb{R}$ is a convex functional.

Proof. (a): The case $\alpha = 0$ is trivial. For $\alpha > 0$ we have

$$\begin{aligned}\varphi'_+(u, \alpha v) &= \lim_{h \searrow 0} \frac{1}{h} (\varphi(u + h\alpha v) - \varphi(u)) \\ &= \alpha \lim_{h \searrow 0} \frac{1}{h\alpha} (\varphi(u + h\alpha v) - \varphi(u)) = \alpha \varphi'_+(u, v).\end{aligned}$$

(b): The representation

$$u + h(v + w) = \frac{1}{2}(u + 2hv) + \frac{1}{2}(u + 2hw),$$

in combination with the convexity of φ , implies

$$\begin{aligned}\varphi'_+(u, v + w) &= \lim_{h \searrow 0} \frac{1}{h} (\varphi(u + h(v + w)) - \varphi(u)) \\ &\leq \lim_{h \searrow 0} \frac{1}{h} \left(\frac{1}{2} \varphi(u + 2hv) + \frac{1}{2} \varphi(u + 2hw) - \varphi(u) \right) \\ &= \lim_{h \searrow 0} \frac{1}{2h} (\varphi(u + 2hv) - \varphi(u)) + \lim_{h \searrow 0} \frac{1}{2h} (\varphi(u + 2hw) - \varphi(u)) \\ &= \varphi'_+(u, v) + \varphi'_+(u, w).\end{aligned}$$

(c): For $u \in \mathcal{F}$, the Gâteaux derivative $\varphi'_+(u, \cdot) : \mathcal{F} \rightarrow \mathbb{R}$ is convex, since

$$\begin{aligned}\varphi'_+(u, \lambda v + (1 - \lambda)w) &\leq \varphi'_+(u, \lambda v) + \varphi'_+(u, (1 - \lambda)w) \\ &= \lambda \varphi'_+(u, v) + (1 - \lambda) \varphi'_+(u, w)\end{aligned}$$

holds for all $\lambda \in [0, 1]$, by using properties (a) and (b). ■

Remark 3.52. By the properties (a) and (b) in Theorem 3.51, we call the functional $\varphi'_+(u, \cdot) : \mathcal{F} \rightarrow \mathbb{R}$ *sublinear*. We can show that the sublinearity of $\varphi'_+(u, \cdot)$, for all $u \in \mathcal{F}$, in combination with the inequality

$$\varphi'_+(u, v - u) \leq \varphi(v) - \varphi(u) \quad \text{for all } u, v \in \mathcal{F},$$

implies the convexity of φ . To see this, we refer to Exercise 3.80. □

Now we show further elementary properties of the Gâteaux derivative.

Theorem 3.53. *Let $\varphi : \mathcal{F} \rightarrow \mathbb{R}$ be a continuous functional. Suppose that for $u, v \in \mathcal{F}$ the Gâteaux derivative $\varphi'_+(u, v)$ exists. Moreover, suppose that $F : \mathbb{R} \rightarrow \mathbb{R}$ has a continuous derivative, i.e., $F \in \mathcal{C}^1(\mathbb{R})$. Then the Gâteaux derivative $(F \circ \varphi)'_+(u, v)$ of the composition $F \circ \varphi : \mathcal{F} \rightarrow \mathbb{R}$ exists at u in direction v , and, moreover, the **chain rule***

$$(F \circ \varphi)'_+(u, v) = F'(\varphi(u)) \cdot \varphi'_+(u, v) \quad (3.34)$$

holds.

Proof. For $x := \varphi(u)$ and $x_h := \varphi(u + hv)$, for $h > 0$, we let

$$G(x_h) := \begin{cases} \frac{F(x_h) - F(x)}{x_h - x} & \text{for } x_h \neq x, \\ F'(x) & \text{for } x_h = x. \end{cases}$$

By the continuity of φ we have $x_h \rightarrow x$ for $h \searrow 0$. Since $F \in \mathcal{C}^1(\mathbb{R})$ this implies

$$F'(x) = \lim_{x_h \rightarrow x} G(x_h) = \lim_{h \searrow 0} G(\varphi(u + hv)) = F'(\varphi(u)).$$

Moreover, we have

$$F(x_h) - F(x) = G(x_h)(x_h - x) \quad \text{for all } h > 0.$$

This finally implies

$$\begin{aligned} (F \circ \varphi)'_+(u, v) &= \lim_{h \searrow 0} \frac{1}{h} (F(\varphi(u + hv)) - F(\varphi(u))) \\ &= \lim_{h \searrow 0} \frac{1}{h} (F(x_h) - F(x)) \\ &= \lim_{h \searrow 0} G(x_h) \cdot \lim_{h \searrow 0} \frac{1}{h} (x_h - x) \\ &= \lim_{h \searrow 0} G(\varphi(u + hv)) \cdot \lim_{h \searrow 0} \frac{1}{h} (\varphi(u + hv) - \varphi(u)) \\ &= F'(\varphi(u)) \cdot \varphi'_+(u, v), \end{aligned}$$

proving both the existence of $(F \circ \varphi)'_+(u, v)$ and the chain rule in (3.34). ■

We now formulate a fundamental sufficient and necessary condition for the characterization of minima for convex functionals.

Theorem 3.54. *Let $\varphi : \mathcal{F} \rightarrow \mathbb{R}$ be a convex functional. Moreover, let $\mathcal{K} \subset \mathcal{F}$ be convex and $u_0 \in \mathcal{K}$. Then the following statements are equivalent.*

- (a) $\varphi(u_0) = \inf_{u \in \mathcal{K}} \varphi(u)$.
- (b) $\varphi'_+(u_0, u - u_0) \geq 0$ for all $u \in \mathcal{K}$.

Proof. (b) $\Rightarrow$ (a): Suppose $\varphi'_+(u_0, u - u_0) \geq 0$ for $u \in \mathcal{K}$. Then we have, due to the monotonicity of the difference quotient $D_{u_0, u - u_0}$ in (3.32), in particular for $h = 1$,

$$0 \leq \varphi'_+(u_0, u - u_0) \leq \varphi(u_0 + (u - u_0)) - \varphi(u_0) = \varphi(u) - \varphi(u_0)$$

and thus $\varphi(u) \geq \varphi(u_0)$.

(a) $\Rightarrow$ (b): Suppose $\varphi(u_0) = \inf_{u \in \mathcal{K}} \varphi(u)$.

Then for $u \in \mathcal{K}$ and small enough $h > 0$ the inequality

$$\frac{1}{h}(\varphi(u_0 + h(u - u_0)) - \varphi(u_0)) \geq 0$$

holds, since, due to convexity of $\mathcal{K}$, we have

$$u_0 + h(u - u_0) = hu + (1 - h)u_0 \in \mathcal{K} \quad \text{for all } h \in (0, 1).$$

This finally implies

$$\varphi'_+(u_0, u - u_0) = \lim_{h \searrow 0} \frac{1}{h}(\varphi(u_0 + h(u - u_0)) - \varphi(u_0)) \geq 0.$$

■

Now we wish to use the conditions from Theorem 3.54 to derive direct characterizations of best approximations to $f \in \mathcal{F}$. To this end, we regard the distance functional $\varphi_f : \mathcal{F} \rightarrow \mathbb{R}$, defined as

$$\varphi_f(v) = \|v - f\| \quad \text{for } v \in \mathcal{F}.$$

Note that the distance functional φ_f is, as a composition between two continuous functionals, namely the translation about f and the norm $\|\cdot\|$, a continuous functional. Moreover, $\varphi_f : \mathcal{F} \rightarrow \mathbb{R}$ is convex. Indeed, for $\lambda \in [0, 1]$ and $v_1, v_2 \in \mathcal{F}$ we have

$$\begin{aligned} \varphi_f(\lambda v_1 + (1 - \lambda)v_2) &= \|\lambda v_1 + (1 - \lambda)v_2 - f\| = \|\lambda(v_1 - f) + (1 - \lambda)(v_2 - f)\| \\ &\leq \lambda\|v_1 - f\| + (1 - \lambda)\|v_2 - f\| = \lambda\varphi_f(v_1) + (1 - \lambda)\varphi_f(v_2). \end{aligned}$$

Therefore, φ_f has a Gâteaux derivative, for which the chain rule (3.34) holds.

Now the direct characterization from Theorem 3.54 can be applied to the distance functional φ_f . This leads us to a corresponding equivalence, which is referred to as *Kolmogorov*¹⁴ *criterion*.

For the Gâteaux derivative of the norm $\varphi = \|\cdot\| : \mathcal{F} \rightarrow \mathbb{R}$ we will henceforth use the notation

$$\varphi'_+(u, v) := \varphi'_+(u, v) \quad \text{for } u, v \in \mathcal{F}.$$

¹⁴ ANDREY NIKOLAEVICH KOLMOGOROV (1903-1987), Russian mathematician

Corollary 3.55. (Kolmogorov criterion).

For $f \in \mathcal{F}$, $\mathcal{S} \subset \mathcal{F}$ convex, and $s^* \in \mathcal{S}$ the following statements are equivalent.

- (a) s^* is a best approximation to f .
- (b) $\|'_+(s^* - f, s - s^*) \geq 0$ for all $s \in \mathcal{S}$.

Proof. Using $\varphi(u) = \|u - f\|$ in Theorem 3.54, we see that $s^* \in \mathcal{S}$ is a best approximation to f , if and only if

$$\begin{aligned} \varphi'_+(s^*, s - s^*) &= \lim_{h \searrow 0} \frac{1}{h} (\varphi(s^* + h(s - s^*)) - \varphi(s^*)) \\ &= \lim_{h \searrow 0} \frac{1}{h} (\|s^* + h(s - s^*) - f\| - \|s^* - f\|) \\ &= \lim_{h \searrow 0} \frac{1}{h} (\|s^* - f + h(s - s^*)\| - \|s^* - f\|) \\ &= \|'_+(s^* - f, s - s^*) \geq 0 \quad \text{for all } s \in \mathcal{S}. \end{aligned}$$

■

Remark 3.56. For proving the implication (b) $\Rightarrow$ (a) in Theorem 3.54 we did not use the convexity of $\mathcal{K}$. Therefore, we can specialize the equivalence in Corollary 3.55 to establish the implication

$$\|'_+(s^* - f, s - s^*) \geq 0 \text{ for all } s \in \mathcal{S} \implies s^* \text{ is best approximation to } f$$

for subsets $\mathcal{S} \subset \mathcal{F}$ that are not necessarily convex. □

Now we use the Gâteaux derivative to prove a characterization concerning the uniqueness of best approximations. To this end, we introduce a sufficient condition, which will be more restrictive than that of (plain) uniqueness.

Definition 3.57. Let $\mathcal{F}$ be a linear space with norm $\|\cdot\|$, $\mathcal{S} \subset \mathcal{F}$ be a subset of $\mathcal{F}$, and $f \in \mathcal{F}$. Then $s^* \in \mathcal{S}$ is said to be the **strongly unique best approximation** to f , if there is a constant $\alpha > 0$ satisfying

$$\|s - f\| - \|s^* - f\| \geq \alpha \|s - s^*\| \quad \text{for all } s \in \mathcal{S}.$$

○

Relying on our previous investigations (especially on Theorem 3.54 and Corollary 3.55), we can characterize the strong uniqueness of best approximations for convex subsets $\mathcal{S} \subset \mathcal{F}$ directly as follows.

Theorem 3.58. Let $\mathcal{F}$ be a linear space with norm $\|\cdot\|$ and $\mathcal{S} \subset \mathcal{F}$ be convex. Moreover, suppose $f \in \mathcal{F}$. Then the following statements are equivalent.

- (a) $s^* \in \mathcal{S}$ is the strongly unique best approximation to f .
- (b) There is one $\alpha > 0$ satisfying $\|'_+(s^* - f, s - s^*) \geq \alpha \|s - s^*\|$ for all $s \in \mathcal{S}$.

Proof. Suppose $f \in \mathcal{F}$ and $s^* \in \mathcal{S}$. For $f \in \mathcal{F}$ we regard the convex distance functional $\varphi : \mathcal{S} \rightarrow [0, \infty)$, defined as $\varphi(s) = \|s - f\|$ for $s \in \mathcal{S}$. Now any element in $\mathcal{S} \setminus \{s^*\}$ can be written as a convex combination of the form

$$s^* + h(s - s^*) = hs + (1 - h)s^* \in \mathcal{S} \quad \text{with } s \in \mathcal{S} \setminus \{s^*\} \text{ and } 1 \geq h > 0.$$

Thereby, we can formulate the strong uniqueness of s^* , for some $\alpha > 0$, as

$$\frac{1}{\|s - s^*\|} \frac{1}{h} (\varphi(s^* + h(s - s^*)) - \varphi(s^*)) \geq \alpha \quad \text{for all } s \in \mathcal{S} \setminus \{s^*\} \text{ and } h > 0.$$

By the monotonicity of the Gâteaux derivative, this is equivalent to

$$\varphi'_+(s^*, s - s^*) = \lim_{h \searrow 0} \frac{1}{h} (\varphi(s^* + h(s - s^*)) - \varphi(s^*)) \geq \alpha \|s - s^*\|$$

for all perturbation directions $s - s^* \in \mathcal{S}$, or, in other words, we have

$$\|'_+(s^* - f, s - s^*) \geq \alpha \|s - s^*\| \quad \text{for all } s \in \mathcal{S}.$$

■

Next, we add an important stability estimate to our discussion on strongly unique best approximations. The following result is due to Freud¹⁵.

Theorem 3.59. (Freud).

For a linear space $\mathcal{F}$ with norm $\|\cdot\|$ and a subset $\mathcal{S} \subset \mathcal{F}$, let $s_f^ \in \mathcal{S}$ be the strongly unique best approximation to $f \in \mathcal{F}$ with constant $\alpha > 0$. Moreover, let $s_g^* \in \mathcal{S}$ be a best approximation to $g \in \mathcal{F}$. Then the stability estimate*

$$\|s_g^* - s_f^*\| \leq \frac{2}{\alpha} \|g - f\|$$

holds.

Proof. By the strong uniqueness of s_f^* , we have the estimate

$$\|s_g^* - f\| - \|s_f^* - f\| \geq \alpha \|s_g^* - s_f^*\|,$$

which further implies the inequalities

$$\begin{aligned} \|s_g^* - s_f^*\| &\leq \frac{1}{\alpha} (\|s_g^* - f\| - \|s_f^* - f\|) \\ &\leq \frac{1}{\alpha} (\|s_g^* - g\| + \|g - f\| - \|s_f^* - f\|) \\ &\leq \frac{1}{\alpha} (\|s_f^* - g\| + \|g - f\| - \|s_f^* - f\|) \\ &\leq \frac{1}{\alpha} (\|s_f^* - f\| + \|f - g\| + \|g - f\| - \|s_f^* - f\|) \\ &= \frac{2}{\alpha} \|g - f\|, \end{aligned}$$

which already proves the stated stability estimate. ■

¹⁵ GÉZA FREUD (1922-1979), Hungarian mathematician

Remark 3.60. If the best approximation $s_g^* \in \mathcal{S}$ is for any $g \in \mathcal{F}$ unique, then the mapping $g \mapsto s_g^*$ is well-defined. Further, due to the Freud theorem, Theorem 3.59, this mapping is continuous at all elements $f \in \mathcal{F}$ which have a strongly unique best approximation $s_f^* \in \mathcal{S}$.

If every $f \in \mathcal{F}$ has a strongly unique best approximation $s_f^* \in \mathcal{S}$, such that their corresponding constants $\alpha_f > 0$ are on $\mathcal{F}$ uniformly bounded away from zero, i.e., if there is some $\alpha_0 > 0$ satisfying $\alpha_f \geq \alpha_0 > 0$ for all $f \in \mathcal{F}$, then the mapping $f \mapsto s_f^*$ is, due to the Freud theorem, Theorem 3.59, by

$$\|s_g^* - s_f^*\| \leq \frac{2}{\alpha_0} \|g - f\| \quad \text{for all } f, g \in \mathcal{F},$$

Lipschitz continuous on $\mathcal{F}$ with *Lipschitz constant* $2/\alpha_0$ (see Definition 6.64). $\square$

Next, we will explicitly compute Gâteaux derivatives for relevant norms $\|\cdot\|$. But first we make the following simple observation.

Remark 3.61. For the Gâteaux derivative of $\varphi(u) = \|u\|$ at $u = 0$, we have

$$\|'_+(0, v) = \lim_{h \searrow 0} \frac{1}{h} (\|0 + hv\| - \|0\|) = \|v\|$$

for any direction $v \in \mathcal{F}$. $\square$

Now we compute Gâteaux derivatives for Euclidean norms.

Theorem 3.62. *The Gâteaux derivative of any Euclidean norm $\|\cdot\|$ is given as*

$$\|'_+(u, v) = \left(\frac{u}{\|u\|}, v \right) \quad \text{for all } u \in \mathcal{F} \setminus \{0\} \text{ and } v \in \mathcal{F}.$$

Proof. Let $\mathcal{F}$ be a Euclidean space with norm $\|\cdot\| = (\cdot, \cdot)^{1/2}$.

For $\varphi(u) = \|u\|$ the chain rule in (3.34) holds in particular for $F(x) = x^2$, i.e.,

$$(\| \|^2)'_+(u, v) = 2\|u\| \cdot \|'_+(u, v) \quad \text{for all } u, v \in \mathcal{F}. \quad (3.35)$$

Moreover, we have

$$\begin{aligned} (\| \|^2)'_+(u, v) &= \lim_{h \searrow 0} \frac{1}{h} (\|u + hv\|^2 - \|u\|^2) \\ &= \lim_{h \searrow 0} \frac{1}{h} (\|u\|^2 + 2h(u, v) + h^2\|v\|^2 - \|u\|^2) \\ &= \lim_{h \searrow 0} \frac{1}{h} (2h(u, v) + h^2\|v\|^2) \\ &= 2(u, v). \end{aligned}$$

This implies, for $u \neq 0$ with (3.35),

$$\|'_+(u, v) = \left(\frac{u}{\|u\|}, v \right) \quad \text{for all } v \in \mathcal{F}.$$

■

The Gâteaux derivative of the absolute value $|\cdot|$ is rather elementary.

Lemma 3.63. *For the absolute-value function $|\cdot| : \mathbb{R} \rightarrow [0, \infty)$, we have*

$$\|'_+(x, y) = y \operatorname{sgn}(x) \quad \text{for all } x \neq 0 \text{ and } y \in \mathbb{R}.$$

Proof. First, note that for $x \neq 0$ we have

$$|x + hy| = |x| + hy \operatorname{sgn}(x) \quad \text{for } h|y| < |x|. \quad (3.36)$$

This implies

$$\|'_+(x, y) = \lim_{h \searrow 0} \frac{1}{h} (|x + hy| - |x|) = \lim_{h \searrow 0} \frac{1}{h} (|x| + hy \operatorname{sgn}(x) - |x|) = y \operatorname{sgn}(x).$$

■

By using the observation (3.36), we can compute the Gâteaux derivatives for all L^p -norms $\|\cdot\|_p$, for $1 \leq p \leq \infty$, on the linear space

$$\mathcal{C}(\Omega) = \{u : \Omega \rightarrow \mathbb{R} \mid u \text{ continuous on } \Omega\}$$

of all continuous functions on a compact domain $\Omega \subset \mathbb{R}^d$, $d \geq 1$. We begin with the maximum norm $\|\cdot\|_\infty$ on $\mathcal{C}(\Omega)$, defined as

$$\|u\|_\infty = \max_{x \in \Omega} |u(x)| \quad \text{for } u \in \mathcal{C}(\Omega).$$

Theorem 3.64. *Let $\Omega \subset \mathbb{R}^d$ be compact. Then, for the Gâteaux derivative of the maximum norm $\|\cdot\| = \|\cdot\|_\infty$ on $\mathcal{C}(\Omega)$, we have*

$$\|'_+(u, v) = \max_{\substack{x \in \Omega \\ |u(x)| = \|u\|_\infty}} v(x) \operatorname{sgn}(u(x))$$

for any $u, v \in \mathcal{C}(\Omega)$, $u \neq 0$.

Proof. Suppose $u \in \mathcal{C}(\Omega)$, $u \neq 0$, and $v \in \mathcal{C}(\Omega)$.

" $\geq$ ": We first show the inequality

$$\|'_+(u, v) \geq \max_{\substack{x \in \Omega \\ |u(x)| = \|u\|_\infty}} v(x) \operatorname{sgn}(u(x)).$$

Suppose $x \in \Omega$ with $|u(x)| = \|u\|_\infty$. Then, by (3.36) we have the inequality

$$\begin{aligned}
\frac{1}{h}(\|u + hv\|_\infty - \|u\|_\infty) &\geq \frac{1}{h}(|u(x) + hv(x)| - |u(x)|) \\
&= \frac{1}{h}(|u(x)| + hv(x) \operatorname{sgn}(u(x)) - |u(x)|) \\
&= v(x) \operatorname{sgn}(u(x))
\end{aligned}$$

for $h|v(x)| < |u(x)|$, which by $h \searrow 0$ already implies the stated inequality.

" $\leq$ ": To verify the inequality

$$\|'_+(u, v) \leq \max_{\substack{x \in \Omega \\ |u(x)| = \|u\|_\infty}} v(x) \operatorname{sgn}(u(x))$$

we regard a strictly monotonically decreasing zero sequence $(h_k)_{k \in \mathbb{N}}$ of positive real numbers, so that $\lim_{k \rightarrow \infty} h_k = 0$. Note that for any element $h_k > 0$ there exists some $x_{h_k} \in \Omega$ satisfying

$$\|u + h_k v\|_\infty = |u(x_{h_k}) + h_k v(x_{h_k})| \quad \text{for } k \in \mathbb{N}.$$

Since Ω is compact, the sequence $(x_{h_k})_{k \in \mathbb{N}}$ has a convergent subsequence $(x_{h_{k_\ell}})_{\ell \in \mathbb{N}} \subset \Omega$ with limit element $\lim_{\ell \rightarrow \infty} x_{h_{k_\ell}} = x \in \Omega$. For $\ell \rightarrow \infty$ we get

$$\|u + h_{k_\ell} v\|_\infty = |u(x_{h_{k_\ell}}) + h_{k_\ell} v(x_{h_{k_\ell}})| \longrightarrow \|u\|_\infty = |u(x)| \quad \text{with } h_{k_\ell} \searrow 0,$$

i.e., every accumulation point of $(x_{h_k})_{k \in \mathbb{N}}$ is an extremum of u in Ω .

Moreover, by (3.36) we obtain the inequality

$$\begin{aligned}
&\frac{1}{h_k}(\|u + h_k v\|_\infty - \|u\|_\infty) \\
&\leq \frac{1}{h_k}(|u(x_{h_k}) + h_k v(x_{h_k})| - |u(x_{h_k})|) \\
&= \frac{1}{h_k}(|u(x_{h_k})| + h_k v(x_{h_k}) \operatorname{sgn}(u(x_{h_k})) - |u(x_{h_k})|) \\
&= v(x_{h_k}) \operatorname{sgn}(u(x_{h_k}))
\end{aligned}$$

for $h_k|v(x_{h_k})| < |u(x_{h_k})|$, whereby

$$\lim_{\ell \rightarrow \infty} \frac{1}{h_{k_\ell}}(\|u + h_{k_\ell} v\|_\infty - \|u\|_\infty) = \|'_+(u, v) \leq v(x) \operatorname{sgn}(u(x)),$$

and where $x \in \Omega$ is an extremum of u in Ω , i.e., $|u(x)| = \|u\|_\infty$. ■

Theorem 3.65. *Let $\Omega \subset \mathbb{R}^d$ be compact and suppose $u, v \in \mathcal{C}(\Omega)$. Then, we have*

$$\|'_+(u, v) = \int_{\Omega_+} v(x) \operatorname{sgn}(u(x)) \, dx + \int_{\Omega_0} |v(x)| \, dx \quad (3.37)$$

for the Gâteaux derivative of the L^1 -norm $\|\cdot\| = \|\cdot\|_1$ on $\mathcal{C}(\Omega)$,

$$\|u\|_1 = \int_{\Omega} |u(x)| \, dx \quad \text{for } u \in \mathcal{C}(\Omega),$$

where $\Omega_0 := \{x \in \Omega \mid u(x) = 0\} \subset \Omega$ and $\Omega_+ := \Omega \setminus \Omega_0$.

Proof. For $u, v \in \mathcal{C}(\Omega)$, we have

$$\begin{aligned} & \frac{1}{h} (\|u + hv\|_1 - \|u\|_1) \\ &= \frac{1}{h} \left(\int_{\Omega} |u(x) + hv(x)| \, dx - \int_{\Omega} |u(x)| \, dx \right) \\ &= \frac{1}{h} \int_{\Omega_+} (|u(x) + hv(x)| - |u(x)|) \, dx + \int_{\Omega_0} |v(x)| \, dx. \end{aligned} \quad (3.38)$$

By $\Omega_h := \{x \in \Omega_+ \mid h \cdot |v(x)| < |u(x)|\} \subset \Omega_+$, for $h > 0$, and with (3.36) we represent the first integral in (3.38) as

$$\begin{aligned} & \frac{1}{h} \int_{\mathbb{R}^d} \chi_{\Omega_+}(x) (|u(x) + hv(x)| - |u(x)|) \, dx \\ &= \int_{\mathbb{R}^d} \chi_{\Omega_h}(x) v(x) \operatorname{sgn}(u(x)) \, dx \\ & \quad + \frac{1}{h} \int_{\mathbb{R}^d} \chi_{\Omega_+ \setminus \Omega_h}(x) (|u(x) + hv(x)| - |u(x)|) \, dx, \end{aligned} \quad (3.39)$$

where for $\Omega \subset \mathbb{R}^d$

$$\chi_{\Omega}(x) = \begin{cases} 1 & \text{for } x \in \Omega, \\ 0 & \text{for } x \notin \Omega, \end{cases}$$

denotes the *indicator function* (i.e., the *characteristic function*) for Ω .

Now we estimate the integral in (3.39) from above by

$$\begin{aligned} & \int_{\mathbb{R}^d} \chi_{\Omega_+ \setminus \Omega_h}(x) (|u(x) + hv(x)| - |u(x)|) \, dx \\ & \leq \int_{\mathbb{R}^d} \chi_{\Omega_+ \setminus \Omega_h}(x) (|u(x)| + h|v(x)| - |u(x)|) \, dx \\ & = h \cdot \int_{\mathbb{R}^d} \chi_{\Omega_+ \setminus \Omega_h}(x) |v(x)| \, dx. \end{aligned} \quad (3.40)$$

Since $\chi_{\Omega_h} \longrightarrow \chi_{\Omega_+}$, or, $\chi_{\Omega_+ \setminus \Omega_h} \longrightarrow 0$, for $h \searrow 0$, the statement in (3.37) follows from the representations (3.38), (3.39) and (3.40). $\blacksquare$

To compute the Gâteaux derivatives for the remaining L^p -norms $\|\cdot\|_p$,

$$\|u\|_p = \left(\int_{\Omega} |u(x)|^p \, dx \right)^{1/p} \quad \text{for } u \in \mathcal{C}(\Omega),$$

for $1 < p < \infty$, we need the following lemma.

Lemma 3.66. *For $1 < p < \infty$, let $\varphi(u) = \|u\|_p$ for $u \in \mathcal{C}(\Omega)$, where $\Omega \subset \mathbb{R}^d$ is assumed to be compact. Then, we have*

$$(\varphi^p)'_+(u, v) = p \int_{\Omega} |u(x)|^{p-1} v(x) \operatorname{sgn}(u(x)) \, dx \quad (3.41)$$

for all $u, v \in \mathcal{C}(\Omega)$, $u \not\equiv 0$.

Proof. For $u, v \in \mathcal{C}(\Omega)$, we have

$$\begin{aligned} & \frac{1}{h}(\varphi^p(u + hv) - \varphi^p(u)) \\ &= \frac{1}{h}(\|u + hv\|_p^p - \|u\|_p^p) = \frac{1}{h} \left(\int_{\Omega} |u(x) + hv(x)|^p dx - \int_{\Omega} |u(x)|^p dx \right) \\ &= \frac{1}{h} \left(\int_{\Omega_+} (|u(x) + hv(x)|^p - |u(x)|^p) dx \right) + h^{p-1} \int_{\Omega_0} |v(x)|^p dx, \quad (3.42) \end{aligned}$$

where $\Omega_0 = \{x \in \Omega \mid u(x) = 0\}$ and $\Omega_+ = \Omega \setminus \Omega_0$.

For $x \in \Omega_h = \{x \in \Omega \mid h \cdot |v(x)| < |u(x)|\} \subset \Omega_+$, where $h > 0$, we have

$$\begin{aligned} |u(x) + hv(x)|^p &= (|u(x)| + hv(x) \operatorname{sgn}(u(x)))^p \\ &= |u(x)|^p + p \cdot |u(x)|^{p-1} \cdot hv(x) \operatorname{sgn}(u(x)) + o(h) \text{ for } h \searrow 0 \end{aligned}$$

by (3.36) and by Taylor¹⁶ expansion of $F(u) = u^p$ at $|u|$.

Thereby, we can split the first integral in (3.42) into the sum

$$\begin{aligned} & \frac{1}{h} \left(\int_{\mathbb{R}^d} \chi_{\Omega_+}(x) (|u(x) + hv(x)|^p - |u(x)|^p) dx \right) \\ &= p \int_{\mathbb{R}^d} \chi_{\Omega_h}(x) |u(x)|^{p-1} v(x) \operatorname{sgn}(u(x)) dx + o(1) \quad (3.43) \end{aligned}$$

$$+ \frac{1}{h} \int_{\mathbb{R}^d} \chi_{\Omega_+ \setminus \Omega_h}(x) (|u(x) + hv(x)|^p - |u(x)|^p) dx. \quad (3.44)$$

Now we estimate the expression in (3.44) from above by

$$\begin{aligned} & \frac{1}{h} \int_{\mathbb{R}^d} \chi_{\Omega_+ \setminus \Omega_h}(x) (|u(x) + hv(x)|^p - |u(x)|^p) dx \\ &\leq \frac{1}{h} \int_{\mathbb{R}^d} \chi_{\Omega_+ \setminus \Omega_h}(x) ((|u(x)| + h|v(x)|)^p - |u(x)|^p) dx \\ &= p \int_{\mathbb{R}^d} \chi_{\Omega_+ \setminus \Omega_h}(x) |u(x)|^{p-1} |v(x)| dx + o(1) \quad \text{for } h \searrow 0. \quad (3.45) \end{aligned}$$

Since $\chi_{\Omega_h} \rightarrow \chi_{\Omega_+}$, or, $\chi_{\Omega_+ \setminus \Omega_h} \rightarrow 0$, for $h \searrow 0$, the stated representation in (3.41) follows from (3.42), (3.43), (3.44), and (3.45). ■

Now we can finally provide the Gâteaux derivatives for the remaining L^p -norms $\|\cdot\|_p$, for $1 < p < \infty$.

Theorem 3.67. *Let $\Omega \subset \mathbb{R}^d$ be compact. Moreover, suppose $1 < p < \infty$. Then, for the Gâteaux derivative of the L^p -norm $\|\cdot\| = \|\cdot\|_p$ on $\mathcal{C}(\Omega)$, we have*

$$\|'_+(u, v) = \frac{1}{\|u\|_p^{p-1}} \int_{\Omega} |u(x)|^{p-1} v(x) \operatorname{sgn}(u(x)) dx$$

for all $u, v \in \mathcal{C}(\Omega)$, $u \neq 0$.

¹⁶ BROOK TAYLOR (1685-1731), English mathematician

Proof. The statement follows from the chain rule (3.34) in Theorem 3.53 with $F(x) = x^p$ in combination with the representation of the Gâteaux derivative $(\varphi^p)'_+$ in Lemma 3.66, whereby

$$\varphi'_+(u, v) = \frac{(\varphi^p)'_+(u, v)}{p\varphi^{p-1}(u)} = \frac{p}{p\|u\|_p^{p-1}} \int_{\Omega} |u(x)|^{p-1} v(x) \operatorname{sgn}(u(x)) \, dx,$$

for $\varphi(u) = \|u\|_p$. ■

3.5 Exercises

Exercise 3.68. Consider approximating the parabola $f(x) = x^2$ on the unit interval $[0, 1]$ by linear functions of the form

$$g_{\xi}(x) = \xi \cdot x \quad \text{for } \xi \in \mathbb{R}$$

with respect to the p -norms $\|\cdot\|_p$, for $p = 1, 2, \infty$, respectively. To this end, first compute the distance function

$$\eta_p(\xi) = \|g_{\xi} - f\|_p.$$

Then, determine the best approximation g_{ξ^*} to f satisfying

$$\|g_{\xi^*} - f\|_p = \inf_{\xi \in \mathbb{R}} \|g_{\xi} - f\|_p$$

along with the minimal distance $\eta_p(\xi^*)$, for each of the three cases $p = 1, 2, \infty$.

Exercise 3.69. Suppose we wish to approximate the identity $f(x) = x$ on the unit interval $[0, 1]$, by an exponential sum of the form

$$p_{\xi}(x) = \xi_1 e^{\xi_2 x} + \xi_3 \quad \text{for } \xi = (\xi_1, \xi_2, \xi_3)^T \in \mathbb{R}^3,$$

and with respect to the maximum norm $\|\cdot\|_{\infty}$.

Show that there is *no* best approximation to f from $\mathcal{S} = \{p_{\xi} \mid \xi \in \mathbb{R}^3\}$.

Hint: Use the parameter sequence $\xi^{(k)} = (k, 1/k, -k)^T$, for $k \in \mathbb{N}$.

Exercise 3.70. Regard the linear space $\mathcal{C}[-\pi, \pi]$, equipped with the norm

$$\|g\| := \|g\|_1 + \|g\|_{\infty} \quad \text{for } g \in \mathcal{C}[-\pi, \pi].$$

Moreover, let $f(x) = x$, for $-\pi \leq x \leq \pi$, and

$$\mathcal{S} = \{\alpha \sin^2(\cdot) \mid \alpha \in \mathbb{R}\} \subset \mathcal{C}[-\pi, \pi].$$

Analyze the existence and uniqueness of the approximation problem

$$\min_{s \in \mathcal{S}} \|s - f\|.$$

Exercise 3.71. Let $\varphi : \mathcal{F} \rightarrow \mathbb{R}$ be a convex functional on a linear space $\mathcal{F}$. Prove the following statements for φ .

- (a) If φ has a (global) maximum on $\mathcal{F}$, then φ is constant.
- (b) A *local* minimum of φ is also a *global* minimum of φ .

Exercise 3.72. Let $(\mathcal{F}, \|\cdot\|)$ be a normed linear space, whose norm $\|\cdot\|$ is *not* strictly convex. Show that there exists an element $f \in \mathcal{F}$, a linear subspace $\mathcal{S} \subset \mathcal{F}$, and *distinct* best approximations $s_1^*, s_2^* \in \mathcal{S}$ to f , $s_1^* \neq s_2^*$, satisfying

$$\eta(f, \mathcal{S}) = \|s_1^* - f\| = \|s_2^* - f\|.$$

Hint: Take for *suitable* $f_1, f_2 \in \mathcal{F}$, $f_1 \neq f_2$, satisfying $\|f_1\| = \|f_2\| = 1$ and $\|f_1 + f_2\| = 2$, the element $f = \frac{1}{2}(f_1 + f_2) \in \mathcal{F}$ and the linear subspace $\mathcal{S} = \{\alpha(f_1 - f_2) \mid \alpha \in \mathbb{R}\} \subset \mathcal{F}$.

Exercise 3.73. Transfer the result of Proposition 3.42 to the case of *odd* functions $f \in \mathcal{C}[-1, 1]$. To this end, formulate and prove a corresponding result for subsets $\mathcal{S} \subset \mathcal{C}[-1, 1]$ that are *invariant under point reflections*, i.e., for any $s(x) \in \mathcal{S}$, we have $-s(-x) \in \mathcal{S}$.

Exercise 3.74. Let $(\mathcal{F}, \|\cdot\|)$ be a normed linear space and $T : \mathcal{F} \rightarrow \mathcal{F}$ be a linear operator which is *isometric* on $\mathcal{F}$, i.e., $\|Tv\| = \|v\|$ for all $v \in \mathcal{F}$. Moreover, let $\mathcal{S} \subset \mathcal{F}$ be a non-empty subset of $\mathcal{F}$ satisfying $T(\mathcal{S}) \subset \mathcal{S}$.

First prove statements (a) and (b), before you analyze question (c).

- (a) If $s^* \in \mathcal{S}$ is a best approximation to $f \in \mathcal{F}$ and $T(\mathcal{S}) = \mathcal{S}$, then $Ts^* \in \mathcal{S}$ is a best approximation to $Tf \in \mathcal{F}$.
- (b) If $f \in \mathcal{F}$ is a fixed point of T in $\mathcal{F}$, i.e., $Tf = f$, and $s^* \in \mathcal{S}$ is a unique best approximation to f , then s^* is a fixed point of T in $\mathcal{S}$.
- (c) Suppose $f \in \mathcal{F}$ is a fixed point of T in $\mathcal{F}$. Moreover, suppose there is *no* fixed point of T in $\mathcal{S}$, which is also a best approximation to f . Can you draw conclusions concerning the uniqueness of best approximation to f ?

Use the results from this exercise to prove Proposition 3.42.

Exercise 3.75. In this exercise, we analyze the existence of *discontinuous* linear functionals φ on $(\mathcal{C}[0, 1], \|\cdot\|_2)$ and on $(\mathcal{C}[0, 1], \|\cdot\|_\infty)$. Give examples, if possible.

- (a) Are there *discontinuous* linear functionals on $(\mathcal{C}[0, 1], \|\cdot\|_2)$?
- (b) Are there *discontinuous* linear functionals on $(\mathcal{C}[0, 1], \|\cdot\|_\infty)$?

Exercise 3.76. Let $a \leq x_0 < \dots < x_n \leq b$ be a sequence of pairwise distinct points in $[a, b] \subset \mathbb{R}$ and $\lambda_0, \dots, \lambda_n \in \mathbb{R}$. Show that the mapping $\varphi : \mathcal{C}[a, b] \rightarrow \mathbb{R}$, defined as

$$\varphi(f) = \sum_{k=0}^n \lambda_k f(x_k) \quad \text{for } f \in \mathcal{C}[a, b],$$

is a continuous linear functional on $(\mathcal{C}[a, b], \|\cdot\|_\infty)$ with operator norm

$$\|\varphi\|_\infty = \sum_{k=0}^n |\lambda_k|.$$

Exercise 3.77. Let $(\mathcal{F}, \|\cdot\|)$ be a normed linear space and $\mathcal{S} \subset \mathcal{F}$ be a finite-dimensional linear subspace of $\mathcal{F}$. Moreover, suppose $f \in \mathcal{F}$.

Prove the following statements on linear functionals from the dual space $\mathcal{F}'$.

- (a) If $\varphi \in \mathcal{F}'$ satisfies $\|\varphi\| \leq 1$ and $\varphi(\mathcal{S}) = 0$, i.e., $\varphi(s) = 0$ for all $s \in \mathcal{S}$, then we have

$$\eta(f, \mathcal{S}) = \inf_{s \in \mathcal{S}} \|s - f\| \geq |\varphi(f)|$$

for the minimal distance $\eta(f, \mathcal{S})$ between f and $\mathcal{S}$.

- (b) There exists one $\varphi \in \mathcal{F}'$ satisfying $\|\varphi\| \leq 1$ and $\varphi(\mathcal{S}) = 0$, such that

$$|\varphi(f)| = \eta(f, \mathcal{S}).$$

If $\eta(f, \mathcal{S}) > 0$, then $\|\varphi\| = 1$.

Exercise 3.78. Consider the linear space $\mathcal{F} = \mathcal{C}([0, 1]^2)$, equipped with the maximum norm $\|\cdot\|_\infty$. Approximate the function

$$f(x, y) = x \cdot y \quad \text{for } (x, y)^T \in [0, 1]^2$$

by a function from the linear approximation space

$$\mathcal{S} = \{s \in \mathcal{F} \mid s(x, y) = s_1(x) + s_2(y) \text{ for } (x, y)^T \in [0, 1]^2 \text{ with } s_1, s_2 \in \mathcal{C}[0, 1]\}.$$

- (a) Construct a linear functional of the form

$$\varphi(g) = \sum_{j=1}^4 \lambda_j g(x_j, y_j) \quad \text{for } g \in \mathcal{F}$$

to estimate the minimal distance $\eta(f, \mathcal{S})$ between f and $\mathcal{S}$, where

$$\eta(f, \mathcal{S}) \geq \frac{1}{4}.$$

(b) Show that

$$s^*(x, y) = \frac{x}{2} + \frac{y}{2} - \frac{1}{4} \quad \text{for } (x, y)^T \in [0, 1]^2$$

is a best approximation to f from $\mathcal{S}$ with respect to $\|\cdot\|_\infty$.

Exercise 3.79. Show that the function $\varphi : \mathbb{R}^2 \rightarrow \mathbb{R}$, defined as

$$\varphi(x, y) = \begin{cases} \frac{xy^2}{x^2+y^4} & \text{for } (x, y) \neq 0 \\ 0 & \text{for } (x, y) = 0 \end{cases} \quad \text{for } (x, y)^T \in \mathbb{R}^2,$$

has a Gâteaux derivative at zero, although φ is *not* continuous at zero.

Exercise 3.80. Let $\mathcal{F}$ be a linear space and $\varphi : \mathcal{F} \rightarrow \mathbb{R}$ a functional on $\mathcal{F}$.

Prove the following statements (related to Remark 3.52).

(a) If φ is convex on $\mathcal{F}$, then the Gâteaux derivative φ'_+ is monotone, i.e.,

$$\varphi'_+(u_1, u_1 - u_2) - \varphi'_+(u_2, u_1 - u_2) \geq 0 \quad \text{for all } u_1, u_2 \in \mathcal{F}.$$

(b) Assume that the Gâteaux derivative $\varphi'_+(u, v)$ exists for all $u, v \in \mathcal{F}$. Moreover, assume that $\varphi'_+(u, \cdot) : \mathcal{F} \rightarrow \mathbb{R}$ is sublinear for all $u \in \mathcal{F}$. If the inequality

$$\varphi'_+(u, v - u) \leq \varphi(v) - \varphi(u) \quad \text{for all } u, v \in \mathcal{F},$$

holds, then φ is convex on $\mathcal{F}$.